



To: Trent Dodson, Executive Director, Kenai Watershed Forum

From: Tony Eskelin, Area Research Biologist, ADF&G Division of Sport Fish, Soldotna

Date: April 8, 2026

Subject: ADF&G Kenai River Chinook salmon research projects and options to direct future funds

ADF&G Division of Sport Fish is providing a summary of research projects being conducted on Kenai River Chinook salmon to Kenai Watershed Forum and stakeholders in the "Save the Kenai King Initiative". The intent is to promote awareness of the current research being conducted and to categorize ways where future funding sources and volunteer efforts could be directed as well as to help identify research gaps or future research needs. There are 7 ADF&G directed research projects on Kenai River Chinook salmon that will be conducted in 2026 (and beyond) that are described below. Ideas to direct potential funding are provided for each project but are not ranked between projects. Should you have any questions regarding these projects, please contact Tony Eskelin, ADF&G at anthony.eskelin@alaska.gov or (907) 262-9368.

Kenai River netting

A standardized inriver netting project is conducted annually to estimate the age, sex, and length (ASL) compositions of early- and late-run Chinook salmon. 60 ft nets split with 2 panels of different mesh sizes are drifted daily for approximately 6 hours in a 1/2 mile zone at approximately RM 8 from May 16 to Aug. 20. Genetic tissue samples are also collected on each captured Chinook salmon to provide the inriver run genetic stock compositions postseason by temporal strata and to evaluate the overlap of tributary and mainstem spawning populations for future escapement goal development. This project has also served as a capture and marking event for various salmon studies and has been used in the past in conjunction with the sonar program (below) to help provide estimates of Chinook salmon abundance of all sizes. Funding from this project comes from Federal Aid through the Dingell-Johnson Act. Additional funding could support research efforts to investigate alternate netting locations to boost early run sample size, reduce mortality, and ensure a representative sample is collected.

Kenai River Chinook salmon sonar

Adaptive resolution imaging sonar (ARIS) units are used to produce inriver abundance estimates of large (>34" TL or 750mm METF) Kenai River early- and late-run Chinook salmon at RM 14. "Large fish" are those Chinook salmon that are larger than any other salmon species and can be accurately assessed by sonar because of the lack of overlap in size from other species. A series of 5 sonars are used to count and measure lengths of fish. These estimates of inriver "large" Chinook salmon passage along with harvest data provide the basis for estimating total large fish run size and spawning escapement. This project is conducted annually from May 16 to approximately August 20 each year and is one of the most important Kenai River Chinook salmon assessment projects. Funding from this project comes from Federal Aid through the Dingell-Johnson Act. Additional funding could support machine AI learning, the purchase of backup sonars, or assess new sonar equipment as technology advances.

Kenai River Chinook salmon fecundity and sex identification

Kenai River late-run Chinook salmon have declined in abundance, mean age at maturity, and mean body size, highlighting the need to quantify stock reproductive potential through improved sex identification and fecundity assessment. Existing monitoring programs depend on accurate sex composition data; however, reliable sex identification using external visual cues becomes increasingly difficult when fish are smaller and in pre-spawning condition. This project will evaluate and develop protocols for noninvasive sex identification and nonlethal

fecundity assessment of Kenai River late-run Chinook salmon. Mature Chinook salmon in spawning condition will be sampled during late summer 2026 and 2027. A cumulative two-year total of approximately 380 fish will be sampled across four Kenai River mainstem spawning areas and selected tributaries. Tissue samples will be genotyped for sex using the Ots_SEXY3-1 marker. Results will be compared to field-verified sex of mature fish to quantify misclassification rates. A Clarius L7 HD3 portable ultrasound unit will be deployed to evaluate the feasibility of gonad-based sex identification for fish in pre-spawning maturity to aid in the development of a fecundity index of the Kenai River stock. The accuracy of each sex identification method will be evaluated by comparing method-assigned sex to field-verified reference sex, with misclassification rates summarized by method and sampling stratum. Results will provide recommended protocols for field sex identification and nonlethal fecundity assessment that can be integrated into ongoing Chinook salmon monitoring programs. Funding from this project comes from ADF&G and a Pacific States Marine Fisheries Commission (PSMFC) grant. Additional funding could support staff to sample tributary spawning fish and assist in the development of a fecundity index.

Kenai River Chinook salmon smolt behavior and survival

Chinook salmon smolt behavior and survival is being studied in the Kenai River and nearshore marine waters of Cook Inlet using acoustic transmitters and hydrophone receivers. This is the first ever study of salmon juveniles using Juvenile Salmon Acoustic Telemetry System (JSATS) technology in the State of Alaska. Beginning in 2025, the first year of this 3-year project was used to learn the technology and determine best methods for tagging and tracking Chinook salmon smolts. In 2025, 287 Chinook salmon smolts were tagged with acoustic transmitters and tracked throughout the drainage downstream of Skilak lake and into Cook Inlet. Continued efforts in 2026 and 2027 will be expanded with additional hydrophones and the use of new, smaller ELAT transmitters which will allow smaller-sized smolt and a more representative sample of the population to be tagged. Fish will be captured primarily with minnow traps and beach seines with the possibility of other trapping devices. This large study is a collaboration between ADF&G and UAF with funding primarily from a NOAA grant, which also supports a UAF graduate student for the marine portion of the project. Additional funding could support future work investigating juvenile Chinook salmon use of tributary streams or how smolts travel through the lakes in the Kenai River watershed.

Kenai River Chinook salmon juvenile late fall and early winter migratory behavior

Beginning in the late summer to early fall, movements of Kenai River Chinook salmon juveniles will be tracked from rearing riparian habitat to unknown overwintering areas. Young of the year parr will be captured throughout the Kenai River drainage and implanted with next generation acoustic transmitters prior to when discharge declines in the fall and when most riparian habitat becomes dewatered and unavailable. Hydrophone receivers will track fish to overwintering areas such as tributaries, lakes, or mainstem channels. After fish locations are found, preferred habitats will be assessed, and critical habitats may be documented for future protection efforts. 2026 will be the first year of a two-year study as part of a graduate research project. Additional funding could support a future second phase of this project, tagging fish in known preferred overwintering habitats to further track their behavior and survival. Another key component of this project is to assess performance of next generation transmitters in salmon, as they have not been previously used in salmon. Results from this assessment will be used to provide feedback to the tag developers. Funding for this project is from ADF&G and a PSMFC grant.

Kenai River Chinook salmon juvenile rearing habitat use and assessment of habitat improvement techniques

Habitat improvement and erosion control techniques will be evaluated to determine their efficacy of improving or providing juvenile Chinook salmon summer rearing habitat. A range of water and habitat variables will be measured to determine which combination is preferred by juvenile Chinook salmon. The intent of this project is to help habitat biologists' ability to design and permit habitat restoration projects and erosion control measures that provide the best rearing conditions possible for juvenile Chinook salmon, ultimately helping increase the number of Chinook salmon that survive the freshwater phase of their life cycle. This project is still being developed and is scheduled for two years beginning in 2026 as a graduate research project through the UAF Cooperative Fish and Wildlife Research Unit in collaboration with ADF&G. Additional funding could provide staff or volunteers for the graduate student to collect habitat variables in the assessments. ADF&G also intends to expand on this project in future years based on what is found with this first phase of this research, which could also be a useful target for outside funding or assistance. Funding from this project is from ADF&G and a PSMFC grant.